Basic Geometry

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Contents

1 Theorems

Theorem 1 (Pythagorean Theorem). *In a right triangle with legs of lengths a and b, and hypotenuse of length c, the following relation holds:*

$$a^2 + b^2 = c^2.$$

Proof. Consider a square with side length $(a + b)$. Inside the square, place four identical right triangles with legs a and b and hypotenuse c, arranged so that their hypotenuses form a smaller square in the center.

The area of the large square is

$$(a + b)^2.$$

This area is also equal to the sum of the areas of the four triangles and the small central square.

Each triangle has area

$$\frac{1}{2}ab,$$

so the four triangles together have area

$$4 \cdot \frac{1}{2}ab = 2ab.$$

The remaining central square has side length c, so its area is

$$c^2.$$

Thus,

$$(a + b)^2 = 2ab + c^2.$$

Expanding the left-hand side gives

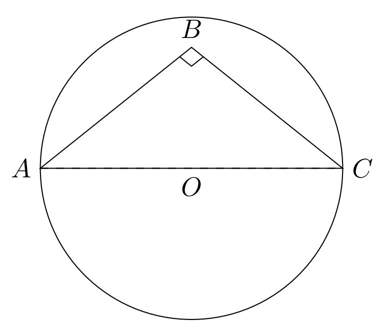
$$a^2 + 2ab + b^2 = 2ab + c^2.$$

Subtracting $2ab$ from both sides yields

$$a^2 + b^2 = c^2.$$

This proves the theorem. □

Theorem 2 (Thales' Theorem). *Let A, B, and C be points on a circle such that AC is a diameter. Then $\angle ABC = 90°$.*



Proof. Let O be the center of the circle. Since AC is a diameter, O is the midpoint of AC.

Because $OA = OB = OC$ (radii of the circle), triangles OAB and OCB are isosceles. Thus,

$$\angle OAB = \angle OBA \quad \text{and} \quad \angle OCB = \angle OBC.$$

Now consider triangle ABC. The angle at B can be written as

$$\angle ABC = \angle OBA + \angle OBC.$$

Using the isosceles triangle relations:

$$\angle OBA = \frac{180^\circ - \angle AOB}{2}, \quad \angle OBC = \frac{180^\circ - \angle BOC}{2}.$$

Since A, O, and C are collinear, we have:

$$\angle AOB + \angle BOC = 180^\circ.$$

Therefore,

$$\angle ABC = \frac{180^\circ - \angle AOB}{2} + \frac{180^\circ - \angle BOC}{2}$$

$$= \frac{360^\circ - (\angle AOB + \angle BOC)}{2} = \frac{360^\circ - 180^\circ}{2} = \boxed{90^\circ}.$$

Hence, $\angle ABC$ is a right angle. □

2 Related Problems

2.1 Problem 1.

First Pythagorean Theorem:

$$12^2 + x^2 = 15^2 \tag{1}$$

$$144 + x^2 = 225 \tag{2}$$

$$x^2 = 81 \tag{3}$$

$$x = 9 \tag{4}$$

Second Pythagorean Theorem:

$$12^2 + y^2 = 13^2 \tag{5}$$

$$144 + y^2 = 169 \tag{6}$$

$$y^2 = 25 \tag{7}$$

$$y = 5 \tag{8}$$

Thus our final answer is $x + y = 14$. □

2.2 Problem 2.

First Pythagorean Theorem:

$$x^2 + (12 - r)^2 = (12 + r)^2 \tag{9}$$

Second Pythagorean Theorem:

$$x^2 + r^2 = (24 - r)^2 \tag{10}$$

Evaluating (9) - (10):

$$(12 - r)^2 - r^2 = (12 + r)^2 - (24 - r)^2 \tag{11}$$

$$(12 - 2r) \cdot 12 = 36 \cdot (2r - 12) \tag{12}$$

Dividing by 12 yields:

$$(12 - 2r) = (2r - 12) \cdot 3 \tag{13}$$

Then dividing by 2:

$$6 - r = 3 \cdot (r - 6) \tag{14}$$

$$6 - r = 3r - 18 \tag{15}$$

Adding 18+r:

$$24 = 4r \tag{16}$$

$$r = 6 \tag{17}$$

□

2.3 Problem 3.

Theorem 3 (Theorem 1.). *For a parallelogram, we have*

$$e^2 + f^2 = a^2 \cdot 2 + b^2 \cdot 2 \tag{18}$$

Proof. First Pythagorean Theorem:

$$(a - x)^2 + m^2 = e^2 \tag{19}$$

Second Pythagorean Theorem:

$$(a + x)^2 + m^2 = f2 \tag{20}$$

Evaulating (19) + (20):

$$a^2 - 2ax + x^2 + m^2 + a^2 + 2ax + x^2 + m^2 = e^2 + f^2 \tag{21}$$

Simplifying and invoking $m^2 + x^2 = b^2$:

$$2a^2 + 2b^2 = e^2 + f^2 \tag{22}$$

□

2.4 Problem 4.

Let the small circle's radius be r.
First Pythagorean Theorem:

$$x^2 + (12 - r)^2 = (12 + r)^2 \tag{23}$$

Second Pythagorean Theorem:

$$r^2 + x^2 = (24 - r)^2 \tag{24}$$

See (1.2) Related Problems: Problem 2. for solution. These are two equivalent problems, under a 90 degree rotation. See figures for clarification.

2.5 Problem 5.

Let's put the circles inside rectangles instead of other circles!
We seek to find the height of the rectangle given the diameter of the small circles.
$r_0 = 12$

2.6 Problem 6.

Another circle-inside-rectangle problem. The rectangle is 60x48, what is the radius of the small circles?